

ANSWER 1. Urban Heat Island Study Using Satellite Data

Introduction

Urban Heat Island (UHI) refers to the phenomenon where urban areas experience higher temperatures than surrounding rural or vegetated areas due to human activities, dense infrastructure, and reduced vegetation. Mapping these variations helps planners design cooling strategies.

Datasets

- **Thermal Satellite Data:** Landsat 8/9 (TIRS bands) for Land Surface Temperature (LST)
- **Multispectral Data:** Sentinel-2 / Landsat for NDVI
- **Land Use/Land Cover (LULC)** maps
- **Administrative boundaries**
- Optional: Weather station data for validation

Methodology

1. Preprocessing

- Convert Digital Numbers (DN) → Radiance → Brightness Temperature
- Perform atmospheric correction
- Clip imagery to study area

2. Land Surface Temperature (LST) Calculation

- Convert thermal band values into temperature
- Apply emissivity correction using NDVI
- Convert Kelvin → Celsius

3. Vegetation Analysis

NDVI formula:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

- High NDVI → dense vegetation → cooler areas
- Low NDVI → built-up → hotter areas

4. Spatial Analysis

- Overlay LST and NDVI maps
- Perform correlation analysis (negative correlation expected)
- Identify hotspots using spatial statistics (e.g., Getis-Ord Gi*)

Expected Outputs

- LST distribution map
- UHI hotspot zones
- NDVI vs LST relationship maps
- Urban cooling priority areas

Conclusion

This study helps urban planners implement **green roofs, parks, and sustainable materials** to reduce temperature and improve livability.

ANSWER 2. Flood Vulnerability Mapping Using DEM

Introduction

Flooding is strongly influenced by terrain. GIS-based terrain analysis helps identify flood-prone zones using elevation and hydrological modeling.

Datasets

- DEM (SRTM / LiDAR)
- Rainfall data
- Soil type
- Land use

- River network

Methodology

1. DEM Preprocessing

- Fill sinks (remove depressions)
- Generate slope and aspect maps

2. Hydrological Modeling

- Flow Direction → determines water movement
- Flow Accumulation → identifies water concentration
- Stream Network Extraction
- Watershed Delineation

3. Factor Analysis

- Low elevation → high flood risk
- Low slope → water stagnation
- High rainfall → increased runoff
- Impervious land → higher flooding

4. Weighted Overlay Analysis

Assign importance weights:

- Elevation (30%)
- Slope (20%)
- Rainfall (25%)
- Land Use (15%)
- Soil (10%)

Combine all layers to create a **Flood Risk Index**

Outputs

- Flood susceptibility map
- Drainage network
- Watershed boundaries
- High-risk zones for planning

Conclusion

Helps in **disaster management, zoning laws, and infrastructure planning.**

ANSWER 3. Pixel Grouping Approaches in Classification

Introduction

Satellite images are raster data made of pixels. Classification converts these pixels into meaningful land cover classes.

Types of Approaches

1. Supervised Classification

- Uses training samples
- Algorithms: Random Forest, SVM, Maximum Likelihood

Working Principle:

Model learns spectral signatures from labeled data.

Advantages

- High accuracy
- Controlled output

Limitations

- Needs training data
- Time-consuming

2. Unsupervised Classification

- No prior labels
- Groups pixels based on similarity

Working Principle:

Clustering (e.g., K-means) groups similar spectral values.

Advantages

- Fast
- No prior data needed

Limitations

- Less accurate
- Needs interpretation

3. Object-Based Image Analysis (OBIA)

- Groups pixels into objects

Working Principle:

Segmentation based on shape, texture, context

Advantages

- Better for high-resolution imagery
- Reduces noise

Limitations

- Complex
- Requires parameter tuning

Applications

- Urban mapping
- Agriculture monitoring
- Forest analysis

ANSWER 4. Surface Characteristics and Urban Temperature

Introduction

Urban temperature patterns are influenced by land surface characteristics like vegetation, built-up areas, and building density.

Key Factors

1. Vegetation Cover

- Reduces temperature via evapotranspiration
- Measured using NDVI

2. Impervious Surfaces

- Asphalt and concrete absorb heat
- Increase temperature significantly

3. Building Density

- Dense areas trap heat
- Reduce airflow

GIS Analysis Methods

- NDVI mapping → vegetation
- LST mapping → temperature
- Overlay analysis
- Regression analysis (NDVI vs LST)

Outputs

- Heat maps

- Correlation maps
- Urban heat risk zones

Conclusion

Supports **green urban planning and climate adaptation strategies.**

ANSWER 5. GIS-Based Plantation Site Selection

Introduction

Selecting plantation sites requires analyzing multiple environmental factors.

Datasets

- DEM (slope)
- Soil type
- Rainfall
- Water sources
- Land use

Methodology

1. Data Preparation

- Standardize projection and resolution

2. Reclassification

Assign suitability scores:

- Gentle slope → high suitability
- Fertile soil → high suitability

3. Weight Assignment

- Soil (30%)

- Water (25%)
- Slope (20%)
- Rainfall (15%)
- Land use (10%)

4. Weighted Overlay

Combine layers to create suitability index

5. Classification

- Highly suitable
- Moderately suitable
- Unsuitable

Outputs

- Plantation suitability map
- Priority zones

Conclusion

Ensures **efficient resource use and higher plantation success rates.**

ANSWER 6. Integration of Spatial Data for Sustainable Urban Development

Introduction

Modern cities generate large spatial datasets that can be integrated for planning and sustainability.

Data Sources

- Satellite imagery
- IoT sensors
- Survey data
- GIS layers

Methods

1. Raster Analysis

- Temperature maps
- Vegetation indices

2. Overlay Analysis

- Combine multiple datasets
- Example: Flood risk + population

3. Spatial Modeling

- Predict urban growth
- Climate risk assessment

Applications

- Smart city planning
- Disaster management
- Climate resilience

Outputs

- Decision-support maps
- Urban dashboards
- Sustainable land-use plans